Multi-Modal Traffic Scenario Generation for Autonomous Driving System Testing Supplementary Material

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1 Supplementary Experiment Results for RQ5: Sensitivity to the Choice of LLMs and Hallucination Errors

Table 1. IE accuracy and textual IR IE accuracy of TrafficComposer using different LLMs in the textual description parser.

LLM Used	Textual IR Acc.(%)	IE Acc.(%)
Llama-3.1-8B-Instruct	71.9(±1.5)	91.1(±1.6)
Llama-3.1-70B-Instruct	73.1(±1.8)	92.9(±1.8)
Llama-3.1-405B-Instruct	76.1(±1.4)	96.5(±1.5)
claude-3-5-sonnet-20240620	77.3(±2.1)	97.0(±2.1)
mistral-large-2407	74.0(±1.8)	94.3(±2.0)
Amazon Titan Text Premier	69.2(±1.4)	90.7(±1.5)
gpt-4o-mini-2024-07-18	74.8(±1.3)	94.0(±1.3)
gpt-4o-2024-05-13	79.2(±1.1)	97.0(±1.2)

Table 1 presents the information extraction (IE) accuracy and the textual IR IE accuracy of TrafficComposer using different LLMs in the textual description parser. The textual IR IE accuracy measures the distance between the textual IR generated by the LLM and the ground truth final traffic IR, using Equation 1 in the paper. We run each experiment three times and report the mean value and margin of error. Across all tested LLMs, TrafficComposer demonstrates robust performance, with average IE accuracy ranging from 90.7% to 97.0%. Although there is a noticeable performance difference between the higher-capacity GPT-4o (gpt-4o-2024-05-13) and the Titan (amazon.titan-text-premier-v1), TrafficComposer achieves acceptable results even with the least effective LLM.

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